

Lecture Note

(Review) Regression Analysis I

- (A) What You Should Know by Now
- (B) Linear Regression Models
- (C) The Expected Value of the OLS Estimators
- (D) The Variance of the OLS Estimators
- (E) Appendix

(A) What You *Should* Know by Now

- Ideally, you need to be familiar with the following concepts.
 - (a) Three data structures
 - (b) Simple linear regression models, Multiple linear regression models
 - (c) Classical Linear Regression Model (CLRM), Assumptions of CLRM, Gauss Markov Assumptions
 - (d) Small (or finite) sample properties of OLS estimators, large sample properties of OLS estimators
 - (e) Calculating estimates and variances, interpreting OLS results, understanding hypothesis testing involved t, F, LM, and χ^2 tests
 - (f) Consequences of violations of Assumptions, Remedies for Assumption violations
- You should spend considerable amount of time on reviewing Chapters 2-6, and Appendices A-C from the textbook. We will use Stata in this course.

(B) Linear Regression Models

Example: Estimating the return to education in a multiple regression model

Econometric model:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \beta_4 married \\ + \beta_5 black + \beta_6 south + \beta_7 urban + u$$

This is an example of a **linear regression model**.

- $\log_e(wage)$ is called the dependent (or **explained**) **variable**, and variables *educ*, *exper*, *tenure*, *married*, *black*, *south*, *urban* are called the **independent** (or explanatory) **variables** (or **regressors**).
- **Definition:** u is the **error term** (or **disturbance**), representing all other omitted variables which affect $\log(wage)$, e.g., ability is excluded because unobservable.
- β_0 is an **intercept parameter**, and $\beta_1, \beta_2, \dots, \beta_7$ are the **slope parameters**.

- Consider a simple regression model that contains only one x variable. Let $\{(x_i, y_i): i = 1, \dots, n\}$ denote a random sample of size n from the population.

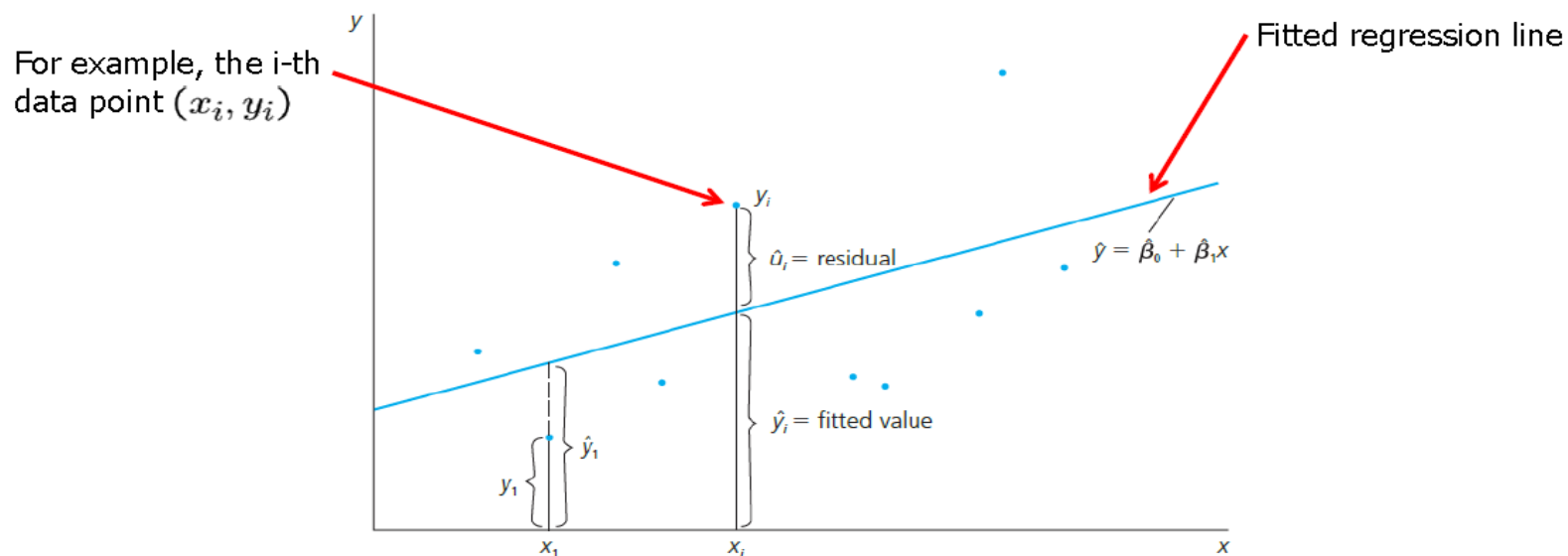
$$y_i = \beta_0 + \beta_1 x_i + u_i, i = 1, 2, \dots, n$$

- Based on the sample $\{(x_i, y_i): i = 1, \dots, n\}$, we can estimate β_0 and β_1 . The $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ is called the sample **regression function**. Both the $\hat{\beta}_0$ and $\hat{\beta}_1$ estimates will be varying across different samples.

- Notations:**

- $\hat{\beta}_0, \hat{\beta}_1$ (“hat”) are the **estimates** of β_0, β_1
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$: $\hat{y}_i$ is the **predicted** (or **fitted**) value of y_i
- $\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$: the **residual** for observation i

Example: Graphical Illustration (How do we find the sample regression line?)



1. The **residual** $\hat{u}_i = y_i - \hat{y}_i$ is observable that is the difference between actual and predicted value of y_i while the error term u is not observable.
2. $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ is called the **sample regression line** which is different from the population regression line. We will never know the population regression line. Consequently, we will never know how close the simple regression function is to the population regression function. Nevertheless, we know we need to minimize the residuals, $\hat{u}_i$ in the OLS estimation.

- **Definition:** $\hat{\beta}_0, \hat{\beta}_1$, the **Ordinary Least Squares (OLS)** estimates of β_0, β_1 , are defined to minimize the **sum of squared residuals (SSR)**,

$$\sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

- The first-order conditions of this optimization problem:

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \dots\dots\dots (1)$$

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1} = \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \dots\dots\dots (2)$$

- The second derivatives can be easily checked. The second derivative test will confirm that the pair of $\hat{\beta}_0$ and $\hat{\beta}_1$ indeed minimizes the sum of squared residuals.

- Solve the equations (1) and (2), and the estimator (or estimates) $\hat{\beta}_0$ is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} ,$$

and the estimator (or estimate) of slope $\hat{\beta}_1$ is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Example: $salary_i = \beta_0 + \beta_1 educ_i + u_i, i = 1, 2, \dots, n$

The parameter β_1 is known as return to education. We can estimate it using the above estimator $\hat{\beta}_1$.

However, there are several technical issues/problems we need to deal with. Can you think of some of them?

- We extend the discussion to the multiple regression models. Let $\{(x_{i1}, \dots, x_{ik}, y_i) : i = 1, \dots, n\}$ denote a random sample of size n from the population: $y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + u_i, \quad i = 1, 2, \dots, n$

Notations:

- (a) $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ (“hat”): the **estimates** of $\beta_0, \beta_1, \dots, \beta_k$
- (b) $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_k x_{ik}$, where $\hat{y}_i$ the **fitted value** or **predicted value** of y_i
- (c) $\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}$, where $\hat{u}_i$ the **residual** for observation i

Question:

How do we use the OLS estimation strategy to estimate the unknown population parameters for the following multiple regression model?

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + u_i, \quad i = 1, 2, \dots, n$$

We use $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$, the Ordinary Least Squares (OLS). And, they are defined to minimize the sum of squared residuals (SSR)

$$\begin{aligned} \min \sum_{i=1}^n \hat{u}_i^2 &= \min \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \min_{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_K x_{ik})^2 \end{aligned}$$

The first-order conditions of this optimization problem:

$$\begin{aligned}
 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\
 \sum_{i=1}^n x_{i1} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\
 \sum_{i=1}^n x_{i2} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\
 &\vdots \\
 \sum_{i=1}^n x_{ik} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0.
 \end{aligned}$$

There are $(k + 1)$ linear equations, and we can solve for the $(k + 1)$ unknowns $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$

$$\hat{\beta} = (X'X)^{-1}X'y \quad \text{Don't need derive}$$

where $\hat{\beta}$ is a $(k + 1) \times 1$ matrix, X is an $n \times (k + 1)$ matrix with rank $(k + 1)$, and y is a $n \times 1$.

- For a particular sample, we can obtain the $\hat{\beta}_j$ estimate. In practice, however, it is almost impossible that the $\hat{\beta}_j$ estimate is equal to the (true) population parameter β_j . (Does it mean it is useless to do the regression analysis?)

(C) The Expected Value of the OLS Estimators

- OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are **random variables**, and they are the functions of a random sample $\{(x_{i1}, \dots, x_{ik}, y_i): i = 1, \dots, n\}$. Given a specific sample, they are fixed. But they vary with different samples drawn from the population.
- The below equation formally states the **population model**, linear in the parameters $\beta_0, \beta_1, \dots, \beta_k$.

- **ASSUMPTION MLR. 1 (LINEAR IN PARAMETERS):** The model in the population can be written as

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u,$$

where $\beta_0, \beta_1, \dots, \beta_k$ are unknown population parameters (constants) of interest, and u is an observable random error.

- **ASSUMPTION MLR. 2 (RANDOM SAMPLING):** We have a random sample of n observations $\{(x_{i1}, \dots, x_{ik}, y_i): i = 1, \dots, n\}$ from the population model described by the above equation.

- **ASSUMPTION MLR. 3 (NO PERFECT COLLINEARITY):** In the sample (and therefore in the population), none of the independent variables is constant, and there are no exact linear relationships among the independent variables.
- **Discussion**
 - (a) The no perfect collinearity assumption concerns only the explanatory variables. If an explanatory variable in the population model is an exact linear combination of the other explanatory variables, then we say the model suffers from **perfect collinearity**, and it cannot be estimated by OLS (redundancy).
 - (b) It is important to note that Assumption MLR.3 *does* allow the explanatory variables to be correlated; they just cannot be *perfectly* correlated.

Example: Consider the relationship between consumption, monthly income, and yearly income as follows:

$$cons = \beta_0 + \beta_1 monthly_inc + \beta_2 yearly_income + u$$

Suppose none of the interviewees receives bonus. This model violates the assumption 3: No perfect collinearity.

Example: Consider the relationship between consumption (*cons*) and income (*inc*):

$$cons = \beta_0 + \beta_1 inc + \beta_2 inc^2 + u$$

The explanatory variables here are correlated, but not perfectly correlated. Therefore, Assumption 3 is not violated.

- **ASSUMPTION MLR. 4 (ZERO CONDITIONAL MEAN):** In a multiple regression model: $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u$, we assume

$$E(u|x_1, \dots, x_k) = E(u) = 0$$

Example: $wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$

In this model, the zero conditional mean assumption implies that other factors affecting *wage* are not related on average to *educ*, *exper* and *tenure*.

Therefore, if we think innate ability is part of u , then we will need average ability levels to be the same across all combinations of education and experience in the working population.

- **Discussion**

- (a) At a minimum, $E(u|x_1, \dots, x_k) = E(u) = 0$ requires that all factors in the unobserved error term be uncorrelated with the explanatory variables.
- (b) When a key variable has been omitted from the equation and absorbed into the error term, $E(u|x_1, \dots, x_k) = E(u) = 0$ may fail since the omitted variable could be correlated with independent variables. In empirical analysis there are always factors that, due to data limitations or ignorance, we will not be able to include. If these factors should be controlled for and they are correlated with one or more of the independent variables, then, $E(u|x_1, \dots, x_k) = E(u) = 0$ will be violated and the **omitted variable** bias will occur.
- (c) When $E(u|x_1, \dots, x_k) = E(u) = 0$ holds, we often say we have **exogenous explanatory variables** $x_1, x_2, \dots, x_K$. If x_j is correlated with u for any reason, then x_j is said to be an endogenous **explanatory variable**.

- **Theorem: Unbiasedness of OLS:** Under Assumptions MLR1 through MLR4.

$$E(\hat{\beta}_j) = \beta_j, j = 1, 2, \dots, k$$

for any values of the population parameter β_j . In other words, the OLS estimators are unbiased estimators of the population parameters.

Example: Consider a simple case of a population model with two explanatory variables:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

If the Assumptions MLR.1 through MLR. 4 hold, the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ are unbiased, $E(\hat{\beta}_j) = \beta_j$, where $j = 0, 1$, and 2.

- Unbiasedness says nothing about the estimate that we obtain for _____ sample. If the sample we obtain is somehow “typical,” then our estimate should be “near” the population value. Unfortunately, it is always possible that we could obtain an unlucky sample that would give us a point estimate far from β_1 , and we can *never* know for sure whether this is the case.
- The unbiasedness of OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ relies on the most important assumption, i.e., the **zero conditional mean assumption**, $E[u|x] = E(u) = 0$ and the other discussed 3 assumptions
- Under these four assumptions, the OLS estimators for the linear regression model are unbiased. The OLS estimators are centered about the true population parameters β_j where $j = 0, 1, 2, \dots, k$. [What does it really mean?]

(D) The Variance of the OLS Estimators

- **ASSUMPTION MLR. 5 (HOMOSKEDASTICITY):**

$$\text{Var}(u|x_1, x_2, \dots, x_k) = \sigma^2.$$

- **Discussion:**

(a) Assumption MLR.5 means that the variance in the error term, u , conditional on the explanatory variables, is the same for all combinations of outcomes of the explanatory variables. If this assumption fails, then the model exhibits heteroskedasticity, just as in the two-variable case.

(b) Assumptions MLR.1 through MLR.5 are collectively known as the **Gauss-Markov** assumptions (for cross-sectional regression).

(c) We can write Assumptions MLR.1 and MLR.4 as

$$E(y|x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_K x_K$$

- **Theorem: Sampling Variance of the OLS Slope Estimations:** Under Assumptions 1 through 5, conditional of the sample values of the explanatory variables,

$$Var(\hat{\beta}_j | x_1, x_2, \dots, x_k) = \frac{\sigma^2}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 (1 - R_j^2)}, \quad j = 1, 2, \dots, k$$

where $SST_j = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$ is the total sample variation in x_j , and R_j^2 is the R-squared from regression x_j on all other explanatory variables and including an intercept.

- **Discussion:**

- (a) This variance formula is invalid in the presence of heteroskedasticity.
- (b) $Var(\hat{\beta}_j)$ depends on the error variance σ^2 , the total variation in x_j , i.e., SST_j , and R_j^2 . A larger variance means a less precise estimator.
- (c) For a given σ^2 and SST_j , the smallest $Var(\hat{\beta}_j)$ is obtained when $R_j^2 = 0$, x_j has zero sample correlation with *all other* explanatory variables.
- (d) High (but not perfect) correlation between two or more of the independent variables is called **multicollinearity**. The case of where R_j^2 is close to one (e.g., 0.9) does not violate Assumption MLR.3.

- Replacing σ^2 with an unbiased estimator $\hat{\sigma}^2$ in the variance $Var(\hat{\beta}_j)$, we obtain unbiased estimators of $Var(\hat{\beta}_j)$. The square root of $\hat{\sigma}^2$, $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ is called the **standard error of the regression (SER)**. Using $\hat{\sigma}$, we can obtain the **standard error** of $\hat{\beta}_j$ to construct the confidence of $\hat{\beta}_j$.

$$se(\hat{\beta}_j) = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 (1 - R_j^2)}} = \frac{\hat{\sigma}}{\sqrt{SST_j (1 - R_j^2)}}$$

- Remark

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n \hat{u}_i^2}{n - k - 1} = \frac{SSR}{n - k - 1}$$

The term $n - k - 1$ is the **degrees of freedom (*df*)** for the general OLS problem with n observations and k independent variables. Since there are $(k + 1)$ parameters in a regression model with k explanatory variables and an intercept, we can write

$$df = n - (k + 1) = (\# \text{ of observations}) - (\# \text{ of estimated parameters})$$

Technically, the division by $n - k - 1$ comes from the fact that the expected value of the sum of squared residuals is $E(SSR) = (n - k - 1) \sigma^2$. Intuitively, we can figure out why the degrees of freedom adjustment is necessary by returning to the first order conditions for the OLS estimators. In obtaining the OLS estimates, $k + 1$ restrictions are imposed on the OLS residuals. This means that, given $n - (k + 1)$ of the residuals, the remaining $k + 1$ residuals are known: there are only $(n - k - 1)$ degrees of freedom in the residuals.

Theorem: (Gauss Markov Theorem)

Under Assumptions 1-5, the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k$ are the best linear unbiased estimator of $\beta_0, \beta_1, \dots, \beta_k$.

(E) Appendix

[A] Cross-sectional data ($y_i, x_i, i=1,2,\dots,n$):

A sample that consists of many individuals (or households or firms or cities or countries, etc.) taken at a same point of time.

(a) Random sampling from the underlying population

(b) Each observation (aka cross-section unit) corresponds to a person or a firm, etc.

(c) Change the order of observations won't change the information/estimation results. (Refer to the point (a))

Example: Cross-sectional Data

TABLE 1.1 A Cross-Sectional Data Set on Wages and Other Individual Characteristics

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

Indicator variables
(1 = yes, 0 = no)

Observation number

Hourly wage

Years of
education

Years of
experience

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[B]Time series data ($y_t, x_t, t=1,2,\dots,T$):

A sample consists of a sequence of observations which are ordered in time

(a) Observing only one entity; it can be a country, a firm or a person, etc.

(b) Track down that particular entity over a period of time, say 43 years. For example, Singapore quarterly GDP from 1980 Q1 -2022 Q4, i.e., 172 quarterly GDP observations and each observation corresponds to the quarter observed

(c) A time series data has a natural temporal ordering. Observations are always dependent across time, or “serial correlated”

Example: Time Series Data

Partial Listing of Data on U.S. Inflation and Unemployment Rates, 1948–1996

Year	Inflation	Unemployment
1948	8.1	3.8
1949	−1.2	5.9
1950	1.3	5.3
1951	7.9	3.3
⋮	⋮	⋮
1994	2.6	6.1
1995	2.8	5.6
1996	3.0	5.4

[C] Panel data (y_{it} , x_{it} , $i=1,2,\dots,n$ and $t=1,2,\dots,T$):

A sample consists of a time series for *each* cross-sectional unit (or entity)

(a) Contains multiple time periods for the cross-sectional units

(b) Contain more observations/information

(c) Control unobserved heterogeneity (fixed effects) that cannot be included in the cross-sectional regression; in other words, we can control something vary across entities but do not vary over time.

Example: Panel Data

TABLE 1.5 A Two-Year Panel Data Set on City Crime Statistics

obsno	city	year	murders	population	unem	police
1	1	1986	5	350,000	8.7	440
2	1	1990	8	359,200	7.2	471
3	2	1986	2	64,300	5.4	75
4	2	1990	1	65,100	5.5	75
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
297	149	1986	10	260,700	9.6	286
298	149	1990	6	245,000	9.8	334
299	150	1986	25	543,000	4.3	520
300	150	1990	32	546,200	5.2	493

Each city has two time series observations

Number of police in 1986

Number of police in 1990

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